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Using a nutrient profile index to assess reclamation strategies in the Athabasca oil sands region of northern Alberta

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Abstract

Land reclamation in the Athabasca oil sands region requires construction of entire soil profiles from materials salvaged during mining. Although much attention has been paid to the limited supply of suitable topsoil materials and their impact on ecosystem recovery, suitable clean subsoil materials are also in limited supply, and their efficient and effective use is an important consideration for land managers in the region. Using data from an oil sands reclamation site in northern Alberta, Canada, we compared soil and foliar nutrients to a wildfire-impacted reference ecosystem with a similarity index. Specifically, we evaluated the similarity of forest floor-mineral mix (FFM) and peat-mineral mix (PM) as topsoil, as well as the effect of different depths of salvaged B and C horizon subsoil with PM on top. All reclamation treatments were planted with jack pine (*Pinus banksiana* Lamb.) and trembling aspen (*Populus tremuloides* Michx.), which were used to examine foliar nutrient concentrations. Individual macronutrient concentrations were different among treatments in total soil nutrients, but differences decreased in soil bioavailable nutrients and disappeared altogether in foliar nutrients. The similarity index revealed that distinct differences existed between treatments, with FFM being the most similar to the wildfire site. It also revealed a potential deficiency in foliar and soil bioavailable Mn on PM, and that increased water content of deeper subsoils had little to no effect. With use of this nutrient profile similarity index, reclamation practitioners may be able to determine if different soil prescriptions lead to higher levels of similarity to natural ecosystems more quickly.

1 | INTRODUCTION

In the Athabasca oil sands region (AOSR) of northern Alberta, surface mining for bitumen results in the complete removal of all soil and vegetation to a depth of 80 m (Rowland, Prescott, Grayston, Quideau, & Bradfield, 2009). The Government of Alberta mandates that land disturbed by surface mining must be reclaimed to an equivalent land capability, and that soil materials must be salvaged and conserved for this purpose during land clearing (AESRD, 2013). Depending on reclamation objectives, conserved soil materials, soil amendments, vegetation planting programs, and follow-up adaptive management can be used to accelerate the reclamation process. The question remains as to how to use these elements to create reclaimed landscapes that function as naturally as possible, and how to quantify that. Many different frameworks for assessing soil quality or function have been proposed, but most rely too heavily on agronomic theory, such

Abbreviations: AOSR, Athabasca oil sands region; AWHC, available water-holding capacity; FFM, forest floor-mineral mix; HSD, honestly significant difference; ICP–OES, inductively coupled plasma optic emission spectroscopy; NPI, nutrient profile index; PC, principal component; PCA, principal component analysis; PM, peat–mineral mix; RIS, raw index score.

as fertility and productivity of monocultures (Mukhopadhyay, Maiti, & Masto, 2014; Ojekanmi & Chang, 2014), and not the theories of ecosystem ecology, which rely on the edatopic grid (relationship between soil water regime and nutrient regime; NRC, 2006). To be clear, we are suggesting that instead of an ecosystem productivity approach, reclamation should take a comparison to natural ecosystem function approach. Therefore, we will develop a nutrient profile index (NPI) for evaluating reclamation success based on similarity to natural sites.

Two topsoil materials are commonly used in the AOSR—forest floor-mineral mix (FFM), which is salvaged from upland forest ecosystems, and peat-mineral mix (PM), which is salvaged from lowland ecosystems (Fung & Macyk, 2000). Peat-mineral mix has a high water-holding capacity and generally contains a large supply of soil organic carbon (C) and nitrogen (N), which is believed to facilitate plant establishment and growth (Macdonald, Quideau, & Landhäusser, 2012; McMillan, Quideau, MacKenzie, & Biryukova, 2007). However, PM tends to lack the biological and substrate legacies that are appropriate for creating an upland forest ecosystem (MacKenzie & Quideau, 2010, 2012), which is the target of most reclamation operations in the AOSR. Conversely, FFM has properties more similar to the soil of upland forest ecosystems, including substrate legacies from prior disturbance events such as wildfires (MacKenzie, Hofstetter, Hatam, & Lanoil, 2014). As for biological legacies, directly placed FFM has viable propagules and microorganisms from upland forest organic layers (Macdonald, Snively, Fair, & Landhäusser, 2015b; Mackenzie & Naeth, 2010; McMillan et al., 2007). Unfortunately, the supply of FFM is very limited in the AOSR, which means that FFM must be used as efficiently as possible during soil rebuilding. Although PM is more abundant than FFM, the government-mandated application depths (≥50 cm; Environment and Sustainable Resource Development, 2013) may deplete the stockpiles of this topsoil for future reclamation.

In addition to the choice and depth of topsoil, placement depth of subsoil material is another area of concern in reclamation prescriptions. Highly compacted subsoil or nonsoil fill materials within rooting depth may limit water availability for reestablishing plant communities by reducing the storage capacity of reclaimed soil (Jung, Duan, House, & Chang, 2014; Naeth, Chanasyk, & Burgers, 2011). Much of the fill material in the AOSR is overburden that was removed during mining, in addition to waste products from bitumen upgrading, such as tailings sand (Kessler, Barbour, van Rees, & Dobchuk, 2010; Naeth et al., 2011). In addition to possessing physical properties undesirable for plant growth, these materials may contain substances that are harmful to plants, such as residual hydrocarbons and/or high concentrations of salts (Kessler et al., 2010; Naeth et al., 2011). Suitable subsoil material, like appropriate topsoil material, is not an infinite resource and has costs associated with collection, transport, storage, and placement. Adequate placement depths and layer thicknesses to reduce impediments to plant growth, while mitigating the potential negative effects of residual hydrocarbons, need to be determined for efficient use of subsoil material.

Core Ideas

- The similarity index analyzes many variables simultaneously rather than individual variables.
- The index suggests overall trends in reclamation success and identifies key drivers.
- Forest floor-mineral mix is more similar to natural sites than peat-mineral mix is.
- For peat-mineral mix treatments, deeper subsoil with additional water capacity had little effect.

One way to explore the importance of and interaction between ecosystem functional variables would be to use principal component analysis (PCA), which makes no assumptions about how much variation will be explained by each component (Brejda, Moorman, Karlen, & Dao, 2000). An index can be developed from the most strongly correlated variables with each axis (Masto, Chhonkar, Singh, & Patra, 2008; Mukhopadhyay et al., 2014). This sort of index is useful for repeated measurements, as it immediately determines which indicators drive differences between sites. Principal component analysis performed on each subsequent sampling could show shifts in the indicators that were primarily responsible for differences between sites over the course of ecosystem development. It can also make use of reference or target ecosystems for trying to determine what the indicators should be like, or in other words to create some target functional parameters. The more similarity to reference sites, the closer the reclaimed sites will appear in the PCA results and index calculation (McCune, Grace, & Urban, 2002; Mukhopadhyay et al., 2014).

This study addressed three specific research questions: (i) Do PM and FFM topsoil function similarly to natural reference sites? (ii) When using PM topsoil, does placing subsoil material to a greater depth improve functional similarity with natural reference sites? and (iii) Can PCA be used to generate an index that clearly and accurately quantifies functional similarity between reclamation and reference sites?

2 | MATERIALS AND METHODS

2.1 | Study area and experimental design

The study was conducted on a reclaimed oil sands mine overburden dump site, ~75 km north of Fort McMurray, AB,

TABLE 1 Mean basic soil characteristics for reclaimed mixed tree sites and benchmark plots at study site and natural reference sites. Means are reported with standard error in parentheses

Soil[a]	Total C	Total N	pH	EC[b]
	g kg^{-1}			dS cm^{-1}
Topsoil/A horizon				
PM-Shallow	241.00 (20.29)b[c]	8.62 (1.71)b	7.09 (0.38)a	1.23×10^{-2} (2.04×10^{-3})a
PM-Medium	321.77 (20.64)a	14.57 (2.89)a	7.54 (0.14)a	4.98×10^{-3} (4.40×10^{-3})bc
PM-Deep	228.77 (23.54)b	10.47 (2.29)ab	7.49 (0.02)a	7.86×10^{-3} (7.80×10^{-4})ab
FFM	24.35 (4.85)c	0.76 (0.17)c	5.88 (0.32)a	9.99×10^{-4} (3.50×10^{-4})c
Reference	21.15 (4.31)c	0.97 (0.32)c	6.45 (1.42)a	2.81×10^{-4} (6.23×10^{-5})c
Upper subsoil/B horizon				
PM-Shallow	4.56 (1.59)ab	0.07 (0.10)b	7.27 (0.59)a	1.77×10^{-3} (1.02×10^{-3})ab
PM-Medium	3.38 (0.58)b	0.01 (0.01)b	7.79 (0.22)a	3.22×10^{-3} (2.06×10^{-4})a
PM-Deep	5.23 (1.72)ab	0.06 (0.07)b	7.56 (0.27)a	1.41×10^{-3} (6.93×10^{-4})ab
FFM	3.96 (0.56)ab	0.01 (0.02)b	6.91 (0.54)a	5.22×10^{-4} (8.98×10^{-5})ab
Reference	7.06 (1.77)a	0.40 (0.11)a	5.79 (0.04)b	2.60×10^{-4} (5.66×10^{-5})b

[a]PM, peat-mineral mix; FFM, forest floor-mineral mix.
[b]EC, electrical conductivity.
[c]Treatments marked with the same letter are not significantly from each other for that property in that horizon.

Canada (57°20′5″ N, 111°32′7″ W). Natural ecosystems of the region are composed of boreal mixedwood forests with major upland tree species of jack pine (*Pinus banksiana* Lamb.), trembling aspen (*Populus tremuloides* Michx.), and white spruce [*Picea glauca* (Moench) Voss], and major lowland species of black spruce [*Picea mariana* (Mill.) Britton, Sterns & Poggenb.], birch (*Betula* spp.), and tamarack [*Larix laricina* (Du Roi) K. Koch] (Natural Regions Committee 2006). Upland soils can be dry and coarse-textured Dystric Eluviated Brunisols (Dystrocrepts) or finer-textured Orthic Gray Luvisols (Haplocryalfs). Lowland soils are often saturated and composed primarily of organic materials, making them Typic Fibrisols (Cryofibrists). (Natural Regions Committee, 2006). This region has a continental climate with mean temperatures of 16.8°C for July and −18.8°C for January and a mean annual precipitation of 455 mm (Environment Canada, 2016).

Samples were collected from the Aurora Soil Capping Study (ASCS) of Syncrude Canada, which is a reclaimed overburden dump constructed and capped in 2012 (for more details see Barber, Bockstette, Christensen, Tallon, & Landhäusser, 2015; Hankin, Karst, & Landhäusser, 2015). The treatments used for this study consisted of 20 cm of topsoil material that was either upland-derived coarse textured FFM or lowland-derived peat mix (PM) (see Table 1 for basic soil properties). Depth of subsoil placement was also tested (see Supplemental Figure S1 for profile details). Subsoils were placed at deep (120 cm), medium (70 cm), or shallow (30 cm) thicknesses (Barber et al., 2015; Hankin et al., 2015). Replicated treatments were planted with jack pine (*Pinus basksianna*) and trembling aspen (*Populus tremuloides*) at 10,000 stems per hectare. Ecosystem recovery from the 2011 Richardson Wildfire event (Pinno & Errington, 2016) began at the same time as growth of planted trees and understory plants on the reclamation treatments allowing for a cohort comparison. This natural reference site was classified as a “b” ecosite, with an aspen and jack pine plant community, and Orthic or Eluviated Dystric Brunisols (Beckingham & Archibald, 1996; Haynes & Naidu, 1998). Three replicated vegetation plots were established at the reference site, separated by 100 m. At all sites, soil samples were collected in August 2015, from the A and B horizon of the reference site, and the topsoil and upper soil of the reclaimed sites (Cumulative Environmental Management Association, 2006). Foliar tissue was collected from both species at each plot, from at least five different seedlings, for a total of ∼10 g dry weight, and only new annual growth was sampled.

2.2 | Soil nutrient analysis

A subsample of soil (∼150 g) was weighed into a mason jar (500 mL), adjusted to 60% of field capacity using deionized water, and incubated at 30°C for 2 wk. Bioavailable nutrients in the soil solution were measured using a pair of ion exchange resin membranes, inserted to ensure good soil contact (Johnson, Verburg, & Arnone, 2005). Upon removal, the membranes were washed with distilled water and extracted with 15 ml of 0.5 M hydrogen chloride (HCl) for analysis of cations and anions. Samples were analyzed for ammonium (NH_4^+), nitrate (NO_3^-), and phosphate (PO_4^{3-}) by the salicylate, cadmium reduction, and Kelowna methods respectively (Kalra & Maynard, 1991; Van Lierop, 1988) on a SmartChem discrete wet chemistry analyzer (Westco Scientific). All other extracted nutrients (K, S, Ca, Mg, Mn, Mo, Al, Fe, Cu, Zn, B, Cd, and Na) were measured by inductively coupled plasma optic emission spectroscopy (ICP–OES) using an iCap 6000 Series (ThermoFisher Scientific).

The rest of the soil samples were air dried, sieved to 2 mm, then ground. A subsample of 0.05 g of each soil was digested with 5.0 ml of 1 M HNO_3 in a Mars microwave digestion system (CEM Corporation) (Baker & Suhr, 1982). Digested samples were then filtered through Whatman No. 42 filter paper and diluted with 25 ml deionized water. Again, nutrient concentrations (Ca, Mg, Mn, Al, Fe, Cu, Mo, Ni, Zn, B, S, P, K, and Na) were determined by ICP–OES (ThermoFisher Scientific). Soil C and N were determined by dry combustion using an ECS 4010 CHNSO analyzer (Costech Analytical Technologies). The pH of the soil was measured by saturated paste using deionized water (Kalra & Maynard, 1991).

2.3 | Foliar nutrient analysis

Foliar tissue samples were oven dried at 40°C and then ground. A 0.03-g subsample from each tree species was digested with 5.0 ml of 1 M HNO_3 in a microwave system (CEM Corporation) (Baker & Suhr, 1982). Digested samples were then filtered through Whatman No. 42 filter paper and diluted with 25 ml of ultrapure water. As with the soil samples, ICP–OES was used to determine the concentrations of foliar nutrients (calcium [Ca], magnesium [Mg], manganese [Mn], iron [Fe], copper [Cu], molybdenum [Mo], nickel [Ni], zinc [Zn], boron [B], sulfur [S], phosphorus [P], potassium [K], and sodium [Na]). Total foliar C and N were determined by dry combustion using an elemental analyzer (Costech Analytical Technologies).

2.4 | Statistical analysis

2.4.1 | Univariate statistics

A one-way ANOVA was performed on the six plant macronutrients (N, P, K, S, Ca, and Mg) (Marschner, 2002), within each nutrient category and each horizon, followed by Tukey's honestly significant difference (HSD) test (Tukey, 1949). All four reclamation treatments and the natural reference sites were compared with each other using R (R Core Development Team, 2015, 2016) and the Agricolae package. A one-way ANOVA was also performed on the heights of the two different tree species (aspen and pine) on each treatment type and the reference sites. This was also followed by Tukey's HSD test to compare treatment means. The available water-holding capacity (AWHC) of each plot was calculated according to a land capability classification system (CEMA, 2006), which uses moisture modifiers based on texture to a depth of 1 m. Mean AWHC for each treatment and the natural reference plots was calculated, and means were compared with a one-way ANOVA and Tukey's HSD test.

2.4.2 | Nutrient profile index

The NPI was derived from PCA performed using the correlation cross-products matrix in PC-ORD version 6.19 (MjM Software Design). As part of the PCA procedure, variables were all relativized by column (i.e., each value is divided by the sum of all values for that variable), and a randomization test was also applied to ensure that the product could not be achieved randomly (McCune et al., 2002). Principal components were considered significant if they had eigenvalues >1.0 and explained $\geq 5.0\%$ of the total variation within the examined data (Masto et al., 2008). Variables retained from each principal component (PC) were multiplied by the percentage of variation that the PC explained including negative correlation coefficients, and these values were summed together to generate a raw index score (RIS) (Masto et al., 2008). To keep individual values at similar weights within the NPI, all RIS were relativized by column (individual RIS divided by the sum of RIS) before being substituted into the equation (see the supplemental materials for the full equation). The RIS of the reference sites were used to determine a mean reference score for determining similarity to treatment sites. Each treatment RIS was then compared with the reference RIS to determine absolute similarity to the target reference site. As the absolute difference from the reference site mean approaches zero, the more similar the plot is to the reference site.

3 | RESULTS

3.1 | Univariate comparisons between treatments

3.1.1 | Tree heights

The reference sites and FFM treatments had significantly taller aspen than the PM-Medium and PM-Deep treatments, whereas the PM-Shallow treatment did not have aspen that were significantly different in height from any other treatment (Figure 1). For pine trees, the only significant difference in height was between FFM and the reference sites (Figure 1).

3.1.2 | Available water-holding capacity

The AWHC of the reference sites was significantly greater than the AWHC of the PM-Shallow treatment (Figure 2). No other treatments were significantly different from each other (Figure 2). The full AWHC calculations according to the Land Capability Classification System (LCCS) (CEMA, 2006) are detailed in Supplemental Table S1.

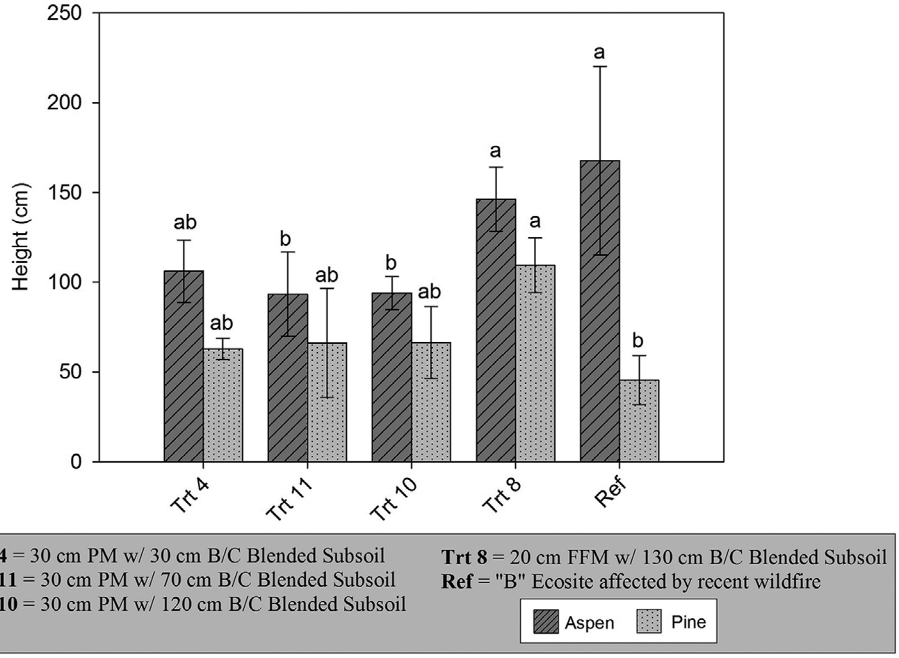

FIGURE 1 Tree heights at the Aurora Soil Capping Study and natural reference (Ref) sites. Treatments (Trt) marked with the same letter do not differ significantly from each other for that species (significance level of $p < .05$). PM, peat-mineral mix; B/C, blended subsoil; FFM, forest floor-mineral mix

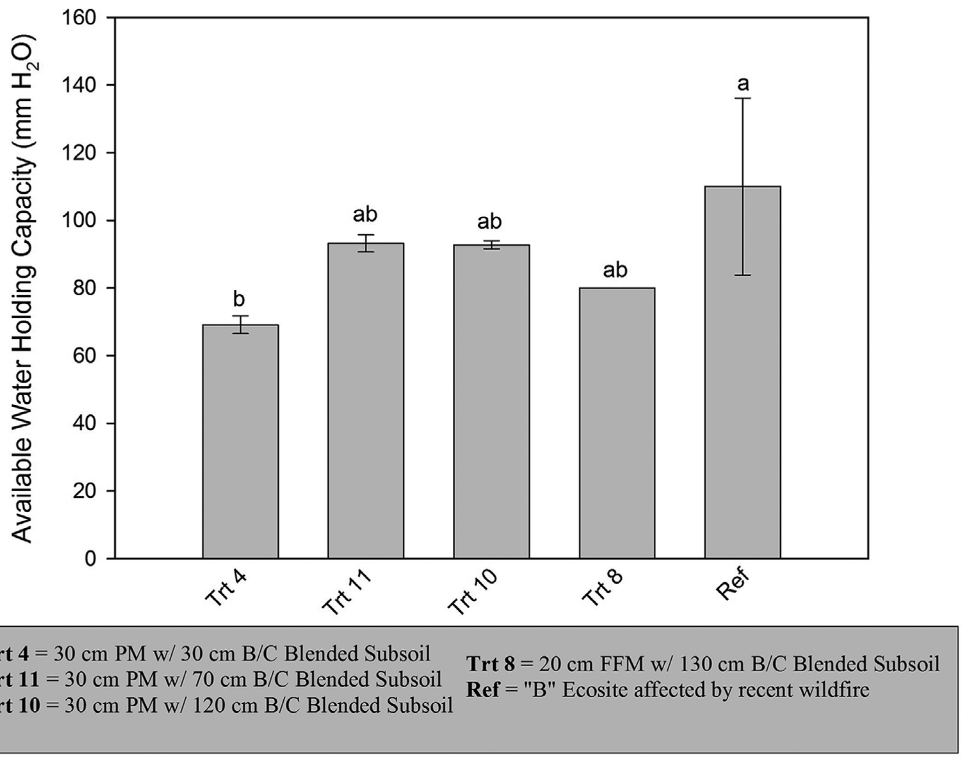

FIGURE 2 Available water-holding capacity in top 100 cm of soil at the Aurora Soil Capping Study and natural reference (Ref) sites. Treatments (Trt) marked with the same letter do not differ significantly from each other (significance level of $p < .05$). PM, peat-mineral mix; B/C, blended subsoil; FFM, forest floor-mineral mix

3.1.3 | Soil total macronutrient concentrations

The soil total nutrients showed significant differences among treatments in the topsoil, with the reference sites and FFM reclamation treatment having lower N, S, Ca, and Mg than the three PM reclamation treatments (Figure 3). Phosphorus was only significantly different between the FFM treatment and PM-Medium treatments with higher P in PM (Figure 3). Potassium was only significantly lower in the FFM treatment compared to the PM-Shallow and PM-Deep treatments (Figure 3).

For soil total nutrients in the upper subsoil, differences were less pronounced, except for the reference sites having greater N and P than all reclamation treatments (Figure 4). The soil total N, S, and Ca levels for the upper subsoil were generally lower than the topsoil soil total nutrient levels for all nutrients and treatments, whereas the P, K, and Mg levels were similar to those found in the topsoil (Figures 3 and 4).

3.1.4 | Soil bioavailable macronutrients

Soil bioavailable nutrients in the topsoil, a much greater range of variability was seen, and no significant differences were observed for N and P, though a pattern of greater P in the FFM and reference sites than in the PM treatments seems to be emerging (Figure 3). The FFM treatment and reference sites both had significantly higher levels of bioavailable K compared with the PM treatments (Figure 3). Other macronutrient differences in topsoil bioavailable nutrients included: S, for which the PM-Shallow treatment had a greater availability than the FFM treatment and reference sites; Ca, for which the PM-Shallow and PM-Deep treatments had a greater availability than the FFM treatment and reference sites; and Mg, for which the PM-Shallow and PM-Deep treatments had a greater availability than the reference sites (Figure 3).

Bioavailable soil nutrients in the upper subsoil showed no significant differences between treatments for N, P, K, and Ca (Figure 4). The PM-Shallow treatment had significantly greater levels of S and Mg than the reference sites (Figure 4). In all nutrients except K, availability was greatly reduced in the upper subsoil, compared with the topsoil (Figures 3 and 4).

3.1.5 | Foliar macronutrient concentrations

Aspen foliar nutrient concentrations differed significantly among treatments for N, for which the PM-Deep and PM-Medium treatments had a greater concentration than the reference sites; P, for which the FFM treatment and reference site had greater concentrations than all PM treatments; and Mg, for which the FFM treatment had a greater concentration than all PM treatments, whereas the reference sites had significantly greater concentrations than the PM-Medium and PM-Deep treatments, but not the PM-Shallow treatment. (Figure 3). No significant differences were found in aspen foliar nutrient concentrations for K, S, or Ca (Figure 3).

For pine foliar nutrient concentrations, significant differences among treatments were observed for P, for which the FFM treatment and reference sites had greater concentrations than all PM treatments; S, for which all reclamation treatments had greater concentrations than the natural reference sites; and Ca, for which all PM treatments had greater concentrations than the FFM treatment and reference sites (Figure 3). No significant differences were found in pine foliar nutrient concentrations for N, K, or Mg (Figure 3).

3.2 | Principal components analysis

Principal component analysis found that there were six principal components with eigenvalues >1.0 and explained ≥5.0% of overall variance (Table 2, Supplemental Table S2). The first PC explained 33.2% of total variation, and the variable with the strongest correlation to the principal component axis was aspen foliar Mn (−92.7%) (Table 2). Seven other variables also had correlations within 10% of aspen foliar Mn's correlation: topsoil soil total S (92.7%), pine foliar Mn (−91.3%), pine foliar Ca (90.0%), topsoil soil total B (89.0%), topsoil soil total C (86.6%), topsoil bioavailable Mg (87.5%), and topsoil bioavailable Ca (85.3%) (Table 2). The second PC explained 12.6% of total variation and the most strongly correlated variable was aspen foliar Fe (−82.2%) (Table 2). No other variables had correlation values within 10% of this value (Supplemental Table S2, Supplemental Figures S2 and S3). The third PC explained 8.4% of total variation and the variable with the strongest correlation was upper subsoil soil total B (−81.3%) (Table 2). Upper subsoil soil total K (−74.2%) was also strongly correlated with the axis of PC3 (Table 2). The fourth PC explained 7.9% of total variation and the variable most strongly correlated to the axis of PC4 was topsoil bioavailable Fe (65.8%) (Table 2). Other variables strongly correlated with the axis of PC4 were topsoil bioavailable Pb (65.2%), topsoil bioavailable Cu (−56.4%), and topsoil bioavailable B (+56.2%) (Table 2). The fifth PC explained 5.6% of total variation, and only topsoil bioavailable Al (−72.9%) was strongly correlated with its axis (3). Finally, PC6 explained 5.2% of total variation, and had pine foliar K (−69.3%) and pine foliar Zn (−68.3%) strongly correlated with the axis (Table 2). The final NPI equation is shown in the supplemental materials.

3.3 | Nutrient profile index

Each individual plot was assessed using the NPI equation (Table 3). The plot that was most similar to the mean

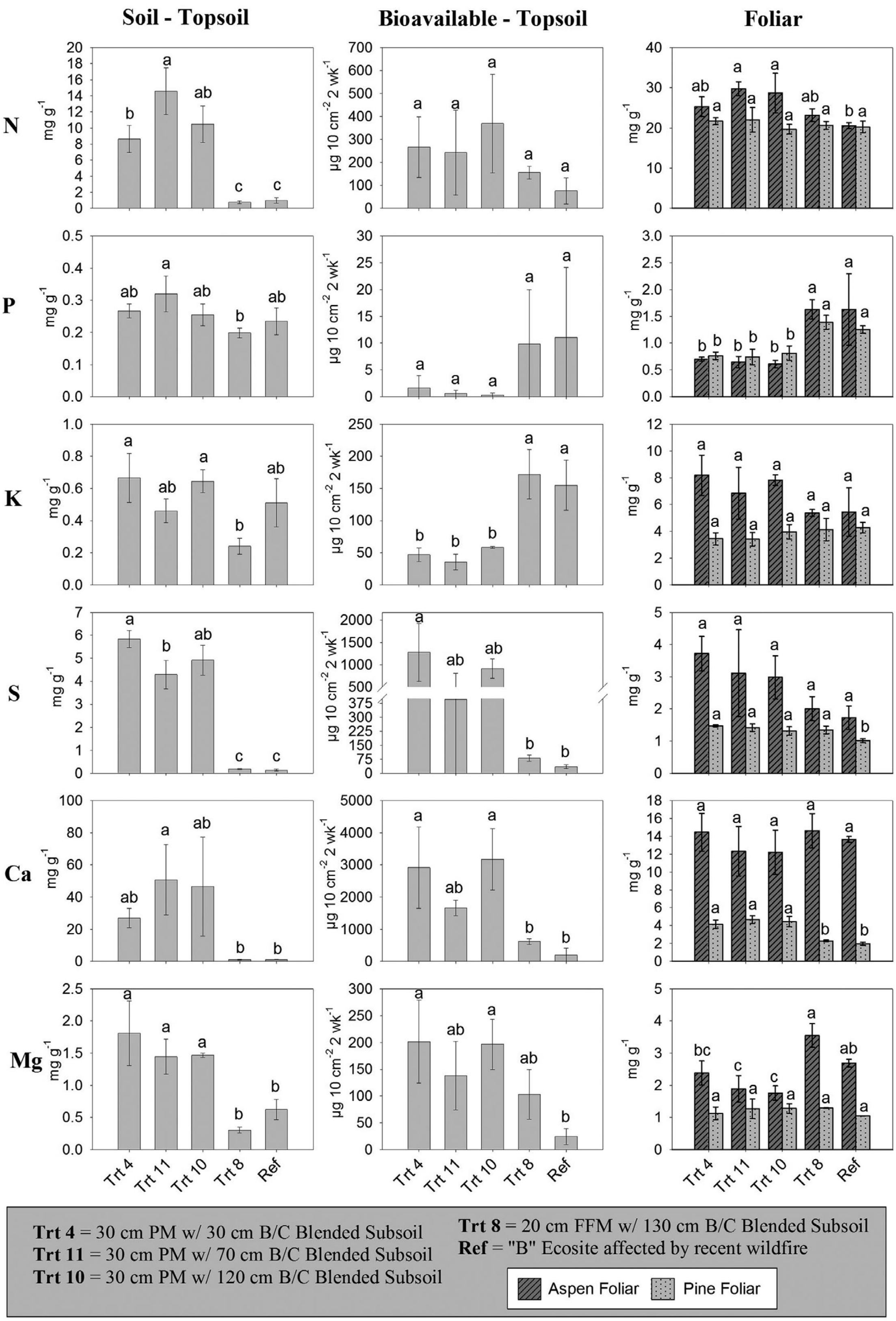

FIGURE 3 Nutrient levels in topsoil soil and bioavailable nutrient pools and foliar nutrient pool from different mixed species reclamation treatments at the Aurora Soil Capping Study (Trt) and natural reference (Ref) sites. Treatments with a non-normal distribution were compared using the Kruskal–Wallis test instead of ANOVA. Treatments marked with the same letter do not differ significantly from each other for that nutrient in that pool (significance level of $p < .05$). Foliar significance levels apply only within each species. PM, peat-mineral mix; B/C, blended subsoil; FFM, forest floor-mineral mix

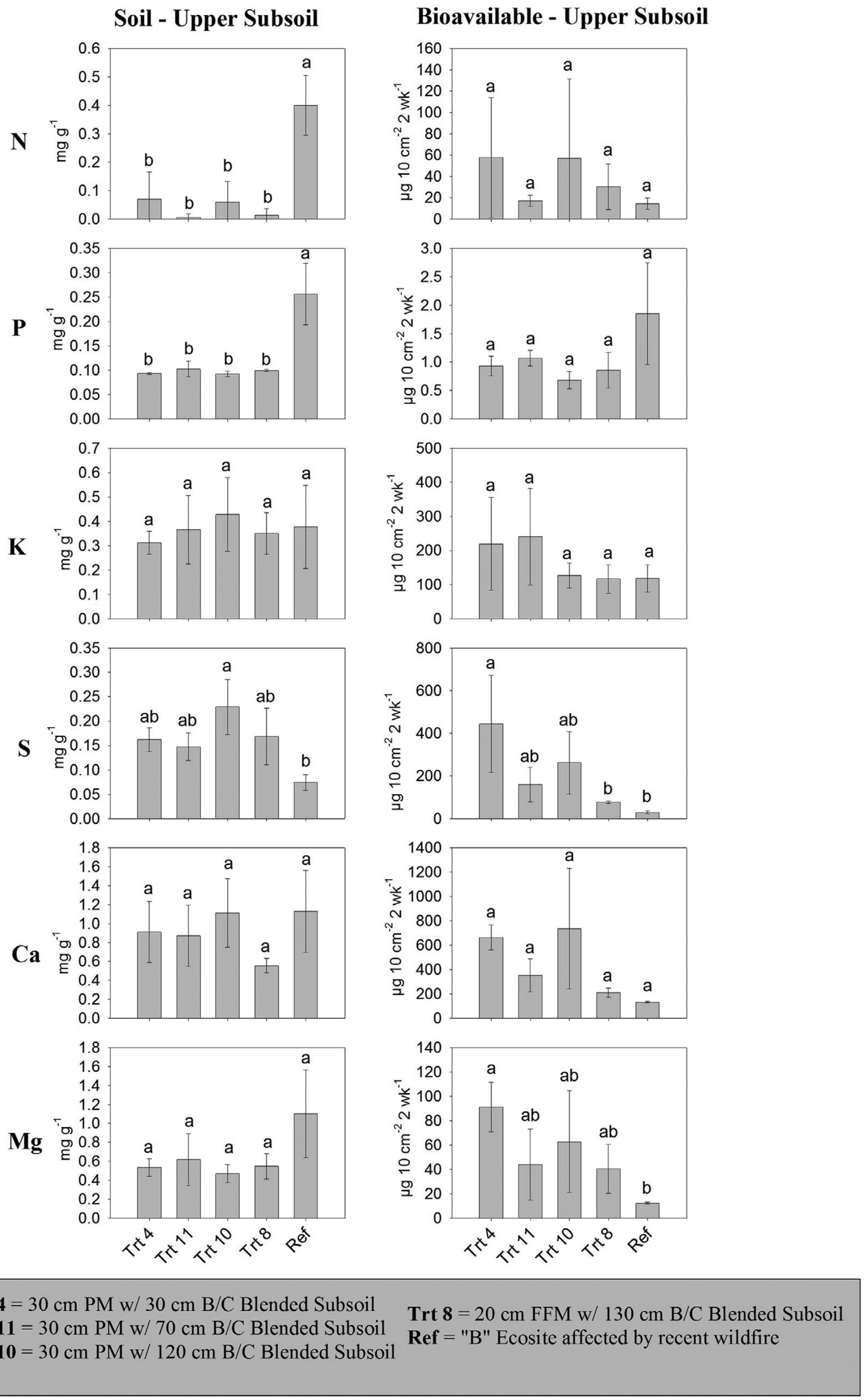

FIGURE 4 Nutrient levels in upper subsoil soil and bioavailable nutrient pools from different mixed species reclamation treatments at the Aurora Soil Capping Study (Trt) and natural reference sites (Ref). Treatments with a non-normal distribution were compared using the Kruskal–Wallis test instead of ANOVA. Treatments marked with the same letter do not differ significantly from each other for that nutrient in that pool (significance level of $p < .05$). PM, peat-mineral mix; B/C, blended subsoil; FFM, forest floor-mineral mix

TABLE 2 Principal component analysis of the correlation matrix of nutrients from soil, bioavailable, and foliar nutrient pools from mixed tree plots at Aurora Soil Capping Study. For each principal component (PC), the bolded values represent the variable with the highest absolute eigenvector (V vector) and every variable with an absolute value within 10% of that value

Parameter	PC1	PC2	PC3	PC4	PC5	PC6
Eigenvalue	28.8	11.0	7.4	6.9	5.7	4.6
Variation (%)	33.2	12.6	8.4	7.9	6.5	5.2
Cumulative variation (%)	33.2	45.8	54.2	62.2	68.7	73.9
Eigenvectors (**V** vectors)						
Environmental physicochemical properties						
Soil total C–topsoil	**0.866**	0.258	−0.100	−0.333	−0.048	−0.002
Foliar nutrients						
Ca–pine	**0.900**	0.171	−0.026	−0.243	−0.106	−0.198
Fe–aspen	0.286	**−0.822**	0.095	0.140	−0.043	−0.039
K–pine	−0.428	−0.094	0.053	0.282	0.087	**−0.693**
Mn–aspen	**−0.927**	−0.110	−0.075	0.107	−0.073	−0.006
Mn–pine	**−0.913**	−0.227	−0.040	0.164	−0.033	−0.030
Zn–pine	0.539	−0.378	−0.166	−0.149	−0.083	**−0.683**
Bioavailable nutrients						
Al–topsoil	0.231	−0.240	0.004	0.427	**−0.729**	0.160
Ca–topsoil	**0.853**	0.181	0.005	0.266	0.050	−0.206
Cu–topsoil	0.037	−0.329	−0.192	**−0.564**	0.173	0.368
Fe–topsoil	0.440	−0.027	−0.323	**0.658**	0.126	0.068
Mg–topsoil	**0.875**	−0.066	−0.010	0.330	−0.082	−0.087
Pb–topsoil	0.220	−0.310	−0.196	**0.652**	−0.279	0.218
Soil nutrients						
B–topsoil	**0.890**	0.356	−0.026	−0.188	−0.08	0.038
B–upper subsoil	0.214	0.217	**−0.813**	0.296	0.115	−0.018
K–upper subsoil	−0.036	0.149	**−0.742**	0.260	0.103	−0.114
S–topsoil	**0.927**	0.275	−0.043	−0.013	0.114	0.119

reference site, plot M-18, was part of the FFM reclamation treatment group (Table 3). The least similar plot to the calculated mean nutrient profile was plot M-25, in the PM-Shallow reclamation treatment group. If the mean NPI is calculated for each treatment group, FFM is most similar to the reference sites, followed by PM-Medium, PM-Deep, and PM-Shallow is the least similar overall (Table 3). A Tukey's HSD test calculated no significant difference for mean NPI between the FFM treatment and the reference sites, whereas the PM reclamation treatments were all significantly different from those two treatment groups, but not from each other (Table 3).

4 | DISCUSSION

4.1 | Generating a similarity index for assessing reclamation success

The NPI establishes the usefulness of multivariate data analysis for determining the most important factors in determining reclamation success, and how the principles of minimum data set analysis can improve representation of differences between sites (Masto et al., 2008). The results of this analysis provide more specific insight into reclamation treatment differences than what is currently recommended (AESRD, 2013). It is possible that refining this index, possibly through use of a standardized scoring function, may further simplify the data and make it easier to interpret (Masto et al., 2008; Mukhopadhyay et al., 2014). However, challenges exist in adopting a scoring function for wildland ecosystem, as there is more inherent variability than in agricultural ecosystems (Ballard & Carter, 1986; Chang, Preston, McCullogh, Weetman, & Barker, 1996; Hu, Hu, Zeng, Tan, & Chang, 2015). Further testing of these techniques on other reclamation sites and natural reference sites, with different vegetation species and methods that incorporate different site total biomass, will help refine and standardize the index and how it is implemented. We believe that the use of nutrient profiles and, in the future, other functional parameters such as soil organic matter quality, C fixation, and organismal diversity will make

TABLE 3 Summary of application of derived nutrient profile index equation to individual plots at study site and natural reference sites. Treatment means were compared to the reference mean to determine relative similarity to the natural reference sites used as a benchmark for recovery from disturbance. Treatments marked with the same letter are not significantly from each other for mean nutrient profile index (NPI)

Treatment[a]	Plot no.	Raw index score	Treatment mean raw index score	Absolute difference from reference mean (NPI)	Mean NPI by treatment
PM-Shallow	M-3[b]	0.1990	0.1774	0.2821	0.2605a[c]
	M-15	0.1265		0.2096	
	M-25	0.2066		0.2897	
PM-Medium	M-7	0.1403	0.1376	0.2234	0.2207a
	M-22	0.1529		0.2359	
	M-26	0.1197		0.2028	
PM-Deep	M-23	0.1931	0.1657	0.2762	0.2488a
	M-30	0.1822		0.2652	
	M-34	0.1212		0.2050	
FFM	M-10	−0.0403	−0.0578	0.0428	0.0252b
	M-18	−0.0632		0.0198	
	M-24	−0.0700		0.0138	
Reference	Ref-1	−0.0715	−0.0831	0.0115	0.0102b
	Ref-2	−0.0984		0.0153	
	Ref-3	−0.0793		0.0038	

[a]PM, peat-mineral mix; FFM, forest floor-mineral mix.
[b]The ASCS contains subplots within each treatment plot that have tree stands of different species compositions, including pure stands of aspen, pine, and spruce, as well as the mixed plots that contain a mixture of all three species (Barber et al., 2015). For this study, only the mixed plots were examined.
[c]Statistically significant differences of the NPI based on ANOVA.

this similarity index more useful than ones relying on edaphic parameters (Mukhopadhyay et al., 2014) or variables auto-correlated to soil organic matter (Ojekanmi & Chang, 2014).

The results of the NPI suggest that the crucial differences between coversoil nutrient profiles are not in bioavaialble N and P, or foliar macronutrients, but rather foliar micronutrients. For example, Mn may be an important nutrient to consider for PM reclamation treatments as evidenced by lower aspen and pine foliar concentrations (Supplemental Table S2). Conversely, total S in the topsoil appears to be in excess at PM sites compared with FFM and natural reference sites, and the same is true of total B in the topsoil, which was strongly correlated with S along the PC1 axis (Supplemental Table S2). Total soil C in the topsoil was also a strong indicator of differences between site types, which is not surprising as the PM treatment has an order of magnitude greater C than the FFM or reference sites. Topsoil bioavailable Mg and Ca, pine foliar Ca, and aspen foliar Fe were also higher in all reclamation treatments than at reference sites (Supplemental Table S2). One implication of these results is that more of any given nutrient, even a necessary macronutrient such as S, Ca, or Mg, is not necessarily better when it comes to reclamation, as this decreases similarity to a site regenerating from natural disturbance, and this may affect overall function. Another implication is that critical micronutrients may play a more important role in ecosystem functioning than commonly monitored macronutrients.

Variables selected from PC3 through PC6 appear to have limited significance with univariate statics (Supplemental Table S2). There may be some concern in emphasizing the impacts of variables, which are associated with PCs that only explain 5.2–8.4% of total variation each, especially when more strongly correlated variables have been discarded (Table 2). We agree that the role of a minimum dataset is not to necessarily to explain the entirety of the differences between designated treatments, but to provide the best possible set of the fewest variables that can still clearly differentiate between groups (Masto et al., 2008). Although the ANOVA results for individual variables may not appear to show significant differences, the NPI does show a clear trend as to which treatments are most similar to the target reference ecosystem.

4.2 | Impact of topsoil material and subsoil depth on similarity

Aspen trees were taller on the FFM treatment and reference sites than on PM treatments. All reclamation treatments had taller pine trees than the reference sites but were not significantly different from each other (Figure 1). Without a control for the impact of the initial nursery conditions on trees outplanted to reclamation sites, or the density of trees on the reference sites, meaningful conclusions about tree height cannot be drawn from comparing reclamation treatments directly to the natural sites. As for differences among reclamation treatments, FFM facilitated the growth of taller trees. There was no significant difference between different subsoil depths on PM treatments for either tree species (Figure 1).

The natural reference sites had the greatest AWHC, but only PM-Shallow had significantly lower AWHC than the reference sites (Figure 2). The PM-Shallow's lower AWHC is due to the soil profiles of PM-Shallow treatment plots having depths of only 60–70 cm compared with the full 100 cm of other treatments (Supplemental Table S1). Although the sandy subsoil material may not provide the greatest water holding capacity, it provides more than the overburden material on which soils are placed. This demonstrates the advantages of having actual soil material occupying the first 100 cm of the constructed soil profile. However, the results of the PCA indicate that this difference in AWHC was never a significant determinant in overall site similarity (Supplemental Table S2). This is surprising as the climate of boreal Alberta has a severe evapotranspirational deficit most years (Environment Canada, 2016), making water the most limiting resource. Additionally, although having deeper soil does provide greater capacity for water and nutrient storage, the most important reservoir appears to be the topsoil layer. Differences in nutrient availability in the upper subsoil layers were much less pronounced between treatments than expected (Figure 4, Supplemental Table S2). The upper subsoil layer was much less likely to provide a significant indicator of differences between sites in the PCA, with only total B and K in the upper subsoil being highly weighted variables (Table 2, Supplemental Table S2).

Differences in univariate statistics were not tied to growth or AWHC parameters at all and had limited significance. Where differences do exist between treatments for individual nutrients, the FFM treatment was generally more similar to reference sites than the PM treatments, including topsoil total N, S, Ca, and Mg; topsoil bioavailable P, K, S, and Ca; aspen foliar P and Mg; and pine foliar P and Ca (Figure 3). Subsoil macronutrients and foliar macronutrients generally did not reflect trends in bioavailable nutrients (Figure 3), and these data alone would not indicate the actual differences in nutrient availability on different sites (Hogberg, Pinno, & MacKenzie, 2019). We suggest that these univariate methods may be appropriate for monoculture work in intensively managed ecosystems but are of limited use in wildland reclamation.

4.3 | Considerations for future work

The Masto et al. (2008) study was conducted on agricultural sites, where soil parameters could adhere to specific prescribed guidelines. This allowed Masto et al. to assign scores to the data, based on whether there was enough of a needed nutrient (e.g., N, P, and K), not too much of a detrimental parameter (e.g., P-fixing capacity), or the parameter values fell into an optimum range (e.g., bulk density). In a wildland ecosystem, it can be argued that there are no appropriate thresholds that can be set in this manner, as the use and cycling of nutrients will not be intensively controlled, nor will nutrients be removed from the ecosystem as harvested biomass. Thus, unlike Masto et al. (2008) and the soil quality index systems based on their work (Mukhopadhyay et al., 2014), we could not standardize each variable to a score out of 1, nor can a value of 1 be a target value for each individual variable. Instead, the mean raw index score for the reference sites was used as the target value for the overall index. This means that when using this system, the target value for the index will be dependent on the number and type of reference sites used. This does allow for the index to be adapted toward multiple types of natural reference sites. For example, if both "a" and "b" ecosite types exist in a region, it would be simple to calculate raw index scores for both site types to compare with reclaimed sites, allowing us to determine different similarity rankings. Whichever reference is selected as the "target" ecosystem, the more similar a reclaimed site is to it, the closer that site's NPI will be to zero.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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